

Factors Affecting Power Consumption for a Given Payload Weight

1. Air Density is governed by three environmental parameters — altitude, temperature, and humidity. Higher altitude reduces air density, forcing the motors to work harder to generate the same thrust. Higher temperatures have a similar effect. Humidity is counterintuitive - more humid air is actually less dense than dry air, which also increases power consumption. All three compounds when testing in hot, humid, high-altitude conditions.

2. Wind Conditions: Both the nature and intensity of wind matter here. Constant wind at a steady speed is manageable - the drone adapts and holds position with relatively predictable power to draw. Sudden gusts, however, force the motors to rapidly decelerate and re-accelerate to maintain position, causing power consumption spikes far beyond what a still hover or steady wind would require.

3. Motor Thermal Efficiency: As flight duration increases, motor temperature rises. Higher operating temperatures reduce motor efficiency, meaning more electrical power is consumed to produce the same mechanical output. This creates a compounding effect - the longer the flight, the worse the efficiency gets.

4. Drone Age and Wear: An older drone with worn motors and degraded propellers will consume more power to lift the same payload compared to a new unit. Component wear directly translates to reduced aerodynamic and electromechanical efficiency.

5. Thrust Blockage: For a given drone model, this is a fixed factor — but it is worth noting. The propellers push air downward, and part of that airflow acts on the drone's own structure, creating a downward pressure on the frame. This is a waste of energy. The extent of this blockage varies across drone models and is a function of the airframe's aerodynamic design.

6. Battery State of Charge: As the battery depletes, the internal resistance increases and the terminal voltage drops. To maintain the same power output, the system draws higher currents, which generates more heat, further reducing efficiency. The result is a non-linear increase in power consumption as the battery drains.

7. Ground Effect: When hovering close to the ground (typically within one rotor diameter of the surface), the downwash from the propellers reflects the ground and creates a high-pressure cushion beneath the drone. This artificially reduces the power needed to hover. Once the drone climbs out of ground effect, this cushion disappears and power consumption increases noticeably — even though the payload weight hasn't changed.